

Question 21

If S has zero content, then $\overline{S}$ has zero content.

Prove

If $\epsilon > 0$, there are rectangles $T_1, T_2, \dots, T_r$ and $T'_1, T'_2, \dots, T'_s$ such that

Since $\cup_{i=1}^r T_i$ is closed, $\overline{S} \subset \cup_{i=1}^r T_i$ if $S \subset \cup_{i=1}^r T_i$.



Question 22

Show that a degenerate rectangle has zero content.

Prve:

Let $R = [a_1, b_1] \times \cdots \times [a_n, b_n]$ and

$$T = [a_1 - \epsilon, b_1 + \epsilon] \times \cdots \times [a_n - \epsilon, b_n + \epsilon], \text{ where } \epsilon > 0.$$

Then $R \subset T$ and, since $a_i = b_i$ for some i , $V(T) \leq 2\epsilon L^{n-1}$, where

$$L = \max \{(b_j - a_j) \mid 1 \leq j \leq n\}.$$



Question 23

$$\int_1^{\infty} \frac{\log x + \sin x}{\sqrt{x}} dx$$

Ans:

Ans

Therefore, $\frac{\log x + \sin x}{\sqrt{x}} > \frac{1}{\sqrt{x}}$ if $x > e^2$. Since $\int_{e^2}^{\infty} \frac{dx}{\sqrt{x}} = \infty$, $\int_{e^2}^{\infty} \frac{\log x + \sin x}{\sqrt{x}} dx = \infty$.

$$\int_1^{\infty} \frac{\log x + \sin x}{\sqrt{x}} dx = \infty$$



Question 24

$$\int_0^{\infty} \frac{1 + \cos^2 x}{\sqrt{1 + x^2}} dx$$

Ans:

$$\frac{1 + \cos^2 x}{\sqrt{1 + x^2}} > \frac{1}{\sqrt{1 + x^2}} = \frac{1}{x \sqrt{1 + 1/x^2}} > \frac{1}{x \sqrt{2}} \text{ if } x > 1. \text{ Since } \int_1^{\infty} \frac{dx}{x} = \infty,$$
$$\int_0^{\infty} \frac{1 + \cos^2 x}{\sqrt{1 + x^2}} dx = \infty$$



Question 25

$$\int_0^{\infty} \frac{4 + \cos x}{(1+x)\sqrt{x}} dx$$

Ans:

$$I = I_1 + I_2 \text{ where } I_1 = \int_0^1 \frac{4 + \cos x}{(1+x)\sqrt{x}} dx \text{ and } I_2 = \int_1^{\infty} \frac{4 + \cos x}{(1+x)\sqrt{x}} dx. \quad 0 < \frac{4 + \cos x}{(1+x)\sqrt{x}} < \frac{5}{\sqrt{x}} \text{ if } x > 0. \text{ Since } \int_0^1 \frac{dx}{\sqrt{x}}, I_1 < \infty. \quad 0 < \frac{4 + \cos x}{(1+x)\sqrt{x}} < \frac{5}{x^{3/2}} \text{ if } x > 1. \text{ Since } \int_1^{\infty} \frac{dx}{x^{3/2}}, I_2 < \infty. \text{ Therefore, } I \text{ converges.}$$



Question 26

The function $f(x, y) = \frac{xy}{x^2 + y^2}$ is defined everywhere in $\mathbb{R}^2$ except at $(0, 0)$.

Does $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$ exist?

Ans:

by letting (x, y) approach $(0, 0)$ along the line $y = x$, we see the functional values

$$f(x, x) = \frac{x^2}{2x^2} = \frac{1}{2} \quad \text{and conclude that the limit is } 1/2.$$

However, if we let (x, y) approach $(0, 0)$ along the line $y = -x$, we see the functional values

$$f(x, -x) = -\frac{x^2}{2x^2} = -\frac{1}{2}$$

and conclude that the limit equals $-1/2$.

In fact, they are both incorrect. What we have shown is that

$$\lim_{x \rightarrow 0} f(x, x) = \frac{1}{2} \quad \text{and} \quad \lim_{x \rightarrow 0} f(x, -x) = -\frac{1}{2}.$$

Since $\lim_{x \rightarrow 0} f(x, x)$ and $\lim_{x \rightarrow 0} f(x, -x)$ must both equal $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$.





If

$$f(x, y, z) = \cos \left(\frac{1}{x^2 + 2y^2 + z^2} \right),$$

then

$$\lim_{|\mathbf{X}| \rightarrow \infty} f(\mathbf{X}) = 1.$$

Ans:

the continuity of $\cos u$ at $u = 0$ implies that for $\varepsilon > 0$
there is a $\delta > 0$ such that

$$|\cos u - 1| < \varepsilon \quad \text{if} \quad |u| < \delta.$$

Since

$$\frac{1}{x^2 + 2y^2 + z^2} \leq \frac{1}{|\mathbf{X}|^2}.$$

It follows that if $|\mathbf{X}| > 1/\sqrt{\delta}$, then

$$\frac{1}{x^2 + 2y^2 + z^2} < \delta.$$

Therefore,

$$|f(\mathbf{X}) - 1| < \varepsilon.$$

Question 28

Suppose that $\mathbf{U}_0$ is in T and is a limit point of T , $\mathbf{G}$ is continuous at $\mathbf{U}_0$, and f is continuous at $\mathbf{X}_0 = \mathbf{G}(\mathbf{U}_0)$. Then $h = f \circ \mathbf{G}$ is continuous at $\mathbf{U}_0$.

Proof: Suppose that $\varepsilon > 0$. Since f is continuous at $\mathbf{X}_0 = \mathbf{G}(\mathbf{U}_0)$, there is an $\varepsilon_1 > 0$ such that

$$|f(\mathbf{X}) - f(\mathbf{G}(\mathbf{U}_0))| < \varepsilon \quad \underline{1}$$

if

$$|\mathbf{X} - \mathbf{G}(\mathbf{U}_0)| < \varepsilon_1 \quad \text{and} \quad \mathbf{X} \in D_f. \quad \underline{2}$$

Since $\mathbf{G}$ is continuous at $\mathbf{U}_0$, there is a $\delta > 0$ such that

$$|\mathbf{G}(\mathbf{U}) - \mathbf{G}(\mathbf{U}_0)| < \varepsilon_1 \quad \text{if} \quad |\mathbf{U} - \mathbf{U}_0| < \delta \quad \text{and} \quad \mathbf{U} \in D_{\mathbf{G}}.$$

By taking $\mathbf{X} = \mathbf{G}(\mathbf{U})$ in (1) and (2)

$$|h(\mathbf{U}) - h(\mathbf{U}_0)| = |f(\mathbf{G}(\mathbf{U})) - f(\mathbf{G}(\mathbf{U}_0))| < \varepsilon$$

if

$$|\mathbf{U} - \mathbf{U}_0| < \delta \quad \text{and} \quad \mathbf{U} \in T.$$

$$\overline{x^2 + 2y^2 + z^2} < \sigma.$$

Therefore,

$$|f(\mathbf{X}) - 1| < \varepsilon.$$



Question 29

Linear Function

A *linear function* is a function of the form $L(\mathbf{X}) = m_1x_1 + m_2x_2 + \cdots + m_nx_n$,

where $m_1, m_2, \dots, m_n$ are constants. From definition of differentiability, f is differentiable at $\mathbf{X}_0$ if and only if there is a linear function L such that $f(\mathbf{X}) - f(\mathbf{X}_0)$ can be approximated so well near $\mathbf{X}_0$ by

$$L(\mathbf{X}) - L(\mathbf{X}_0) = L(\mathbf{X} - \mathbf{X}_0)$$

that

$$f(\mathbf{X}) - f(\mathbf{X}_0) = L(\mathbf{X} - \mathbf{X}_0) + E(\mathbf{X})(|\mathbf{X} - \mathbf{X}_0|),$$

where

$$\lim_{\mathbf{X} \rightarrow \mathbf{X}_0} E(\mathbf{X}) = 0.$$



Question 30

If

$$f(x, y) = \frac{x^2 + y^2}{x - y},$$

Solution:

$$f_x(x, y) = \frac{2x}{x - y} - \frac{x^2 + y^2}{(x - y)^2}$$

$$f_y(x, y) = \frac{2y}{x - y} + \frac{x^2 + y^2}{(x - y)^2}.$$

Since f_x and f_y are continuous on

$$S = \{(x, y) : x \neq y\}$$

f is continuously differentiable on S .



Question 31

Absolute integrability

We say that f is *absolutely integrable* on $[a, b)$ if f is locally integrable on $[a, b)$ and $\int_a^b |f(x)| dx < \infty$. In this case we also say that $\int_a^b f(x) dx$ *converges absolutely* or is *absolutely convergent*.

Conditional convergence

An oscillatory function may be integrable, but not absolutely integrable, on $[a, b)$, as the next example shows. We then say that f is *conditionally* integrable on $[a, b)$, and that $\int_a^b f(x) dx$ *converges conditionally*.



Question 32

Dirichlet's Test

Suppose that f is continuous and its antiderivative $F(x) = \int_a^x f(t) dt$ is bounded on $[a, \hat{b})$.

Let g' be absolutely integrable on $[a, b)$, and suppose that

$\lim_{x \rightarrow b-} g(x) = 0$. Then $\int_a^b f(x)g(x) dx$ converges.

Differentiable Surfaces

A differentiable surface S in $\mathbb{R}^n$ ($n > 1$) is the image of a compact subset D of $\mathbb{R}^m$, where $m < n$, under a continuously differentiable transformation $\mathbf{G} : \mathbb{R}^m \rightarrow \mathbb{R}^n$. If $m = 1$, S is also called a differentiable curve.



Question 33

$$\int_1^{\infty} x^q \sin x \, dx$$

diverges if $q > 0$, but none of the other tests that we have studied so far implies this. It is not enough to argue that the integrand does not approach zero as $x \rightarrow \infty$ (a common mistake), since this does not imply divergence. To see that the integral diverges, we observe that if it converged for some $q > 0$, then $F(x) = \int_1^x x^q \sin x \, dx$ would be bounded on $[1, \infty)$.

We could let

$$f(x) = x^q \sin x \quad \text{and} \quad g(x) = x^{-q}$$

in Dirichlet's test and conclude that

$$\int_1^{\infty} \sin x \, dx$$

also converges. This is false.



Question 34

Upper Integrals

If f is bounded on a rectangle R in $\mathbb{R}^n$ and $\mathbf{P} = \{R_1, R_2, \dots, R_k\}$ is a partition of R .

Let

$$M_j = \sup_{\mathbf{X} \in R_j} f(\mathbf{X}),$$

The upper sum of f over $\mathbf{P}$ is

$$S(\mathbf{P}) = \sum_{j=1}^k M_j V(R_j).$$

The upper integral of f over R , denoted by $\overline{\int}_R f(\mathbf{X}) d\mathbf{X}$,

is the infimum of all upper sums.

Lower Integrals:

If f is bounded on a rectangle R in $\mathbb{R}^n$ and $\mathbf{P} = \{R_1, R_2, \dots, R_k\}$ is a partition of R .

$$m_j = \inf_{\mathbf{X} \in R_j} f(\mathbf{X}).$$

The lower sum of f over $\mathbf{P}$ is

$$s(\mathbf{P}) = \sum_{j=1}^k m_j V(R_j).$$

The lower integral of f over R , denoted by $\underline{\int}_R f(\mathbf{X}) d\mathbf{X}$,

is the supremum of all lower sums.



Question 35

Any bounded set E with only finitely many limit points has zero content.

To see this, we first observe that if E has no limit points, then it must be finite, by the Bolzano–Weierstrass theorem, and therefore must have zero content.

Now suppose that the limit points of E are $\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_m$. Let $R_1, R_2, \dots, R_m$ be rectangles such that $\mathbf{X}_i \in R_i^0$ and

$$V(R_i) < \frac{\varepsilon}{2m}, \quad 1 \leq i \leq m. \quad (3.30)$$

The set of points of E that are not in $\cup_{j=1}^m R_j$ has no limit points (why?) and, being bounded, must be finite (again by the Bolzano–Weierstrass theorem).

If this set contains p points, then it can be covered by rectangles $R'_1, R'_2, \dots, R'_p$ with

$$V(R'_j) < \frac{\varepsilon}{2p}, \quad 1 \leq j \leq p. \quad (3.31)$$

Now,

$$E \subset \left(\bigcup_{i=1}^m R_i \right) \cup \left(\bigcup_{j=1}^p R'_j \right)$$

From (3.30) and (3.31),

$$\sum_{i=1}^m V(R_i) + \sum_{j=1}^p V(R'_j) < \varepsilon.$$



Question 36

Example: The circle

$\{(x, y) : x^2 + y^2 = 9\}$ is a differentiable curve in $\mathbb{R}^2$.

Ans:

Since it is the image of $D = [0, 2\pi]$ under the continuously differentiable transformation $\mathbf{G} : \mathbb{R} \rightarrow \mathbb{R}^2$ defined by

$$\mathbf{X} = \mathbf{G}(\theta) = \begin{bmatrix} 3 \cos \theta \\ 3 \sin \theta \end{bmatrix}.$$

5

Example: The sphere

$\{(x, y, z) : x^2 + y^2 + z^2 = 4\}$ is a differentiable surface in $\mathbb{R}^3$.

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Ans:

Since it is the image of $D = \{(\theta, \phi) : 0 \leq \theta \leq 2\pi, -\pi/2 \leq \phi \leq \pi/2\}$

under the continuously differentiable transformation $\mathbf{G} : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ defined by

$$\mathbf{X} = \mathbf{G}(\theta, \phi) = \begin{bmatrix} 2 \cos \theta \cos \phi \\ 2 \sin \theta \cos \phi \\ 2 \sin \phi \end{bmatrix}.$$



Question 37

Iterated Integrals I_1

assume that f is continuous on $R = [a, b] \times [c, d]$.

Then, for each y in $[c, d]$, $f(x, y)$ is continuous with respect to x on $[a, b]$, so the integral

$$F(y) = \int_a^b f(x, y) dx$$

exists.

Moreover, the uniform continuity of f on R implies that F is continuous and therefore integrable on $[c, d]$.

We say that

$$I_1 = \int_c^d F(y) dy = \int_c^d \left(\int_a^b f(x, y) dx \right) dy$$

is an *iterated integral* of f over R .

which we usually write as

$$I_1 = \int_c^d dy \int_a^b f(x, y) dx.$$

Iterated Integrals I_2

Another iterated integral can be defined by writing

$$G(x) = \int_c^d f(x, y) dy, \quad a \leq x \leq b,$$

We say that

$$I_2 = \int_a^b G(x) dx = \int_a^b \left(\int_c^d f(x, y) dy \right) dx,$$

which we usually write as

$$I_2 = \int_a^b dx \int_c^d f(x, y) dy.$$



Question 38

The function $f(x) = (1-x)^{-p}$ is locally integrable on $[0, 1)$ and, if $p \neq 1$ and $0 < c < 1$,

$$\int_0^c (1-x)^{-p} dx = \frac{(1-x)^{-p+1}}{p-1} \Big|_0^c = \frac{(1-c)^{-p+1} - 1}{p-1}.$$

Hence,

$$\lim_{c \rightarrow 1-} \int_0^c (1-x)^{-p} dx = \begin{cases} (1-p)^{-1}, & p < 1, \\ \infty, & p > 1. \end{cases}$$

For $p = 1$,

$$\lim_{c \rightarrow 1-} \int_0^c (1-x)^{-1} dx = -\lim_{c \rightarrow 1-} \log(1-c) = \infty.$$

Hence,

$$\int_0^1 (1-x)^{-p} dx = \begin{cases} (1-p)^{-1}, & p < 1, \\ \infty, & p \geq 1. \end{cases}$$



Question 39

The function $f(x) = x^{-p}$ is locally integrable on $[1, \infty)$ and, if $p \neq 1$ and $c > 1$,

$$\int_1^c x^{-p} dx = \left. \frac{x^{-p+1}}{-p+1} \right|_1^c = \frac{c^{-p+1} - 1}{-p+1}.$$

Hence,

$$\lim_{c \rightarrow \infty} \int_1^c x^{-p} dx = \begin{cases} (p-1)^{-1}, & p > 1, \\ \infty, & p < 1. \end{cases}$$

For $p = 1$,

$$\lim_{c \rightarrow \infty} \int_1^c x^{-1} dx = \lim_{c \rightarrow \infty} \log c = \infty.$$

Hence,

$$\int_1^\infty x^{-p} dx = \begin{cases} (p-1)^{-1}, & p > 1, \\ \infty, & p \leq 1. \end{cases}$$



Question 40

The function $f(x) = \cos x$ is locally integrable on $[0, \infty)$

$$\lim_{c \rightarrow \infty} \int_0^c \cos x \, dx = \lim_{c \rightarrow \infty} \sin c$$

does not exist; thus, $\int_0^\infty \cos x \, dx$ diverges, but not to $\pm\infty$.

Question

The function $f(x) = \log x$ is locally integrable on $(0, 1]$, but unbounded as $x \rightarrow 0+$. Since

$$\lim_{c \rightarrow 0+} \int_c^1 \log x \, dx = \lim_{c \rightarrow 0+} (x \log x - x) \Big|_c^1 = -1 - \lim_{c \rightarrow 0+} (c \log c - c) = -1,$$

$$\int_0^1 \log x \, dx = -1.$$



Question 41

Let $f(x, y) = x + y$ and $R = [0, 1] \times [1, 2]$.

Ans:

$$F(y) = \int_0^1 f(x, y) dx = \int_0^1 (x + y) dx = \frac{1}{2} + y$$

$$I_1 = \int_1^2 F(y) dy = \int_1^2 \left(\frac{1}{2} + y \right) dy = \left(\frac{y}{2} + \frac{y^2}{2} \right) \Big|_1^2 = 2.$$

Also,

$$G(x) = \int_1^2 (x + y) dy = \left(xy + \frac{y^2}{2} \right) \Big|_{y=1}^2 = x + \frac{3}{2},$$

$$I_2 = \int_0^1 G(x) dx = \int_0^1 \left(x + \frac{3}{2} \right) dx = \left(\frac{x^2}{2} + \frac{3x}{2} \right) \Big|_0^1 = 2.$$



Question 42

Theorem: Suppose that f is integrable on $R = [a, b] \times [c, d]$ and $F(y) = \int_a^b f(x, y) dx$ exists for each y in $[c, d]$.

Ans:

Then F is integrable on $[c, d]$, and

$$\int_c^d F(y) dy = \int_R f(x, y) d(x, y);$$

that is,

$$\int_c^d dy \int_a^b f(x, y) dx = \int_R f(x, y) d(x, y).$$



Proof: Write

$$\mathbf{X} - \mathbf{Z} = (\mathbf{X} - \mathbf{Y}) + (\mathbf{Y} - \mathbf{Z}),$$

and apply triangle inequality with $\mathbf{X}$ and $\mathbf{Y}$ replaced by $\mathbf{X} - \mathbf{Y}$ and $\mathbf{Y} - \mathbf{Z}$.

Corollary: If $\mathbf{X}$ and $\mathbf{Y}$ are in $\mathbb{R}^n$, then

$$|\mathbf{X} - \mathbf{Y}| \geq ||\mathbf{X}| - |\mathbf{Y}||.$$

Proof: Since

$$\mathbf{X} = \mathbf{Y} + (\mathbf{X} - \mathbf{Y}),$$

Triangle inequality implies that

$$|\mathbf{X}| \leq |\mathbf{Y}| + |\mathbf{X} - \mathbf{Y}|,$$

which is equivalent to $|\mathbf{X}| - |\mathbf{Y}| \leq |\mathbf{X} - \mathbf{Y}|$.

Interchanging $\mathbf{X}$ and $\mathbf{Y}$ yields

$$|\mathbf{Y}| - |\mathbf{X}| \leq |\mathbf{Y} - \mathbf{X}|.$$

Since $|\mathbf{X} - \mathbf{Y}| = |\mathbf{Y} - \mathbf{X}|$, the last two inequalities imply the stated conclusion.

Corollary: If $\mathbf{X}$, $\mathbf{Y}$, and $\mathbf{Z}$ are in $\mathbb{R}^n$, then

$$|\mathbf{X} - \mathbf{Z}| \leq |\mathbf{X} - \mathbf{Y}| + |\mathbf{Y} - \mathbf{Z}|.$$

Proof: Write

$$\mathbf{X} - \mathbf{Z} = (\mathbf{X} - \mathbf{Y}) + (\mathbf{Y} - \mathbf{Z}),$$

and apply triangle inequality with $\mathbf{X}$ and $\mathbf{Y}$ replaced by $\mathbf{X} - \mathbf{Y}$ and $\mathbf{Y} - \mathbf{Z}$.

2.15 Bounded Functions

The definitions of *bounded above*, *bounded below*, and *bounded* on a set S are the same for functions of n variables as for functions of one variable, as are the definitions of *supremum* and *infimum* of a function on a set S .

Example: Let $\Phi = (\phi_1, \phi_2, \phi_3)$ and

$$f(x, y, z) = 3xyz + 2x^2 + z^2.$$

Then

$$\begin{aligned}h(t) &= f(x + t\phi_1, y + t\phi_2, z + t\phi_3), \\&= 3(x + t\phi_1)(y + t\phi_2)(z + t\phi_3) + 2(x + t\phi_1)^2 \\&\quad + (z + t\phi_3)^2.\end{aligned}$$

$$h(t) = 3(x + t\phi_1)(y + t\phi_2)(z + t\phi_3) + 2(x + t\phi_1)^2 + (z + t\phi_3)^2$$

Then we have

$$\begin{aligned}h'(t) &= 3\phi_1(y + t\phi_2)(z + t\phi_3) + 3\phi_2(x + t\phi_1)(z + t\phi_3) \\&\quad + 3\phi_3(x + t\phi_1)(y + t\phi_2) + 4\phi_1(x + t\phi_1) \\&\quad + 2\phi_3(z + t\phi_3).\end{aligned}$$

Therefore,

$$\frac{\partial f(\mathbf{X})}{\partial \Phi} = h'(0) = (3yz + 4x)\phi_1 + 3xz\phi_2 + (3xy + 2z)\phi_3.$$

Example: Show that the following function $f(x, y) = x^2 + 2xy$, is differentiable at any point (x_0, y_0) .

$$\begin{aligned}
 f(x, y) - f(x_0, y_0) &= x^2 + 2xy - x_0^2 - 2x_0y_0 \\
 &= x^2 - x_0^2 + 2(xy - x_0y_0) \\
 &= (x - x_0)(x + x_0) + 2(xy - x_0y_0) \\
 &\quad + 2(x_0y - x_0y_0) \\
 &= (x + x_0 + 2y)(x - x_0) + 2x_0(y - y_0) \\
 &= 2(x_0 + y_0)(x - x_0) + 2x_0(y - y_0) \\
 &\quad + (x - x_0)(x - x_0 + 2y - 2y_0) \\
 &= m_1(x - x_0) + m_2(y - y_0) + (x - x_0)(x - x_0 + 2y - 2y_0),
 \end{aligned}$$

where

$$m_1 = 2(x_0 + y_0) = f_x(x_0, y_0) \quad \text{and} \quad m_2 = 2x_0 = f_y(x_0, y_0). \quad (2.45)$$

Therefore,

$$\begin{aligned}
 &\frac{|f(x, y) - f(x_0, y_0) - m_1(x - x_0) - m_2(y - y_0)|}{|\mathbf{X} - \mathbf{X}_0|} \\
 &= \frac{|x - x_0|(x - x_0 + 2(y - y_0))|}{|\mathbf{X} - \mathbf{X}_0|} \\
 &\leq \sqrt{5}|\mathbf{X} - \mathbf{X}_0|,
 \end{aligned}$$

by Schwarz's inequality. This implies that

$$\lim_{\mathbf{X} \rightarrow \mathbf{X}_0} \frac{f(x, y) - f(x_0, y_0) - m_1(x - x_0) - m_2(y - y_0)}{|\mathbf{X} - \mathbf{X}_0|} = 0,$$

so f is differentiable at (x_0, y_0) .

2.7.1 Polygonal Path

If $\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_k$ are points in $\mathbb{R}^n$

Let L_i is the line segment from $\mathbf{X}_i$ to $\mathbf{X}_{i+1}$, $1 \leq i \leq k-1$, we say that $L_1, L_2, \dots, L_{k-1}$ form a *polygonal path* from $\mathbf{X}_1$ to $\mathbf{X}_k$.

We say that $\mathbf{X}_1$ and $\mathbf{X}_k$ are *connected* by the polygonal path.

2.8 Polygonally Connected Set

A set S is *polygonally connected* if every pair of points in S can be connected by a polygonal path lying entirely in S .

Moments of inertia.

1 Solid Circular Cylinder.

$$I = \frac{1}{2} Ma^2$$

2 Hollow Circular Cylinder.

$$I = Ma^2$$

3 Solid Sphere.

$$I = \frac{2}{5} Ma^2$$

4 Rectangular Plate

$$I = \frac{1}{12} M(a^2 + b^2)$$

5 Thin Rod

$$I = \frac{1}{12} Ma^2$$

6 Triangular Lamina

$$I = \frac{1}{6} Mh^2$$

7 Right Circular Cone

$$I = \frac{3}{10} Ma^2$$

2.16 Directional Derivative

Let Φ be a unit vector and $\mathbf{X}$ a point in $\mathbb{R}^n$.

The directional derivative of f at $\mathbf{X}$ in the direction of Φ is defined by

$$\frac{\partial f(\mathbf{X})}{\partial \Phi} = \lim_{t \rightarrow 0} \frac{f(\mathbf{X} + t\Phi) - f(\mathbf{X})}{t}$$

if the limit exists.

2.16. Directional Derivative

97

That is, $\partial f(\mathbf{X})/\partial \Phi$ is the ordinary derivative of the function

$$h(t) = f(\mathbf{X} + t\Phi)$$

at $t = 0$, if $h'(0)$ exists.

3.9 Upper and Lower Integrals

If f is bounded on a rectangle R in $\mathbb{R}^n$ and $\mathbf{P} = \{R_1, R_2, \dots, R_k\}$ is a partition of R .

Let

$$M_j = \sup_{\mathbf{X} \in R_j} f(\mathbf{X}), \quad m_j = \inf_{\mathbf{X} \in R_j} f(\mathbf{X}).$$

The *upper sum* of f over $\mathbf{P}$ is

$$S(\mathbf{P}) = \sum_{j=1}^k M_j V(R_j).$$

The *upper integral* of f over R , denoted by

$$\overline{\int}_R f(\mathbf{X}) \, d\mathbf{X},$$

is the infimum of all upper sums.

Upper and Lower Integrals: The *lower sum* of f over $\mathbf{P}$ is

$$s(\mathbf{P}) = \sum_{j=1}^k m_j V(R_j).$$

The *lower integral* of f over R , denoted by

$$\underline{\int}_R f(\mathbf{X}) \, d\mathbf{X},$$

is the supremum of all lower sums.

Example: let

$$f(x, y) = \begin{cases} (x - y)^2 \sin \frac{1}{x-y}, & x \neq y, \\ 0, & x = y. \end{cases}$$

Then

$$f_x(x, y) = 2(x - y) \sin \frac{1}{x - y} - \cos \frac{1}{x - y}, \quad x \neq y,$$

and

$$f_x(x, x) = \lim_{h \rightarrow 0} \frac{f(x + h, x) - f(x, x)}{h} = \lim_{h \rightarrow 0} \frac{h^2 \sin(1/h) - 0}{h} = 0,$$

so f_x exists for all (x, y) , but is not continuous on the line $y = x$.

Example: Let

$$f(x, y) = \begin{cases} (x - y)^2 \sin \frac{1}{x-y}, & x \neq y, \\ 0, & x = y. \end{cases}$$

The same is true of f_y , since

$$f_y(x, y) = -2(x - y) \sin \frac{1}{x - y} + \cos \frac{1}{x - y}, \quad x \neq y,$$

and

$$f_y(x, x) = \lim_{k \rightarrow 0} \frac{f(x, x + k) - f(x, x)}{k} = \lim_{k \rightarrow 0} \frac{k^2 \sin(-1/k) - 0}{k} = 0.$$

Now,

$$\begin{aligned} & \frac{f(x, y) - f(0, 0) - f_x(0, 0)x - f_y(0, 0)y}{\sqrt{x^2 + y^2}} \\ &= \begin{cases} \frac{(x-y)^2}{\sqrt{x^2+y^2}} \sin \frac{1}{x-y}, & x \neq y, \\ 0, & x = y, \end{cases} \end{aligned}$$

and Schwarz's inequality implies that

$$\left| \frac{(x - y)^2}{\sqrt{x^2 + y^2}} \sin \frac{1}{x - y} \right| \leq \frac{2(x^2 + y^2)}{\sqrt{x^2 + y^2}} = 2\sqrt{x^2 + y^2}, \quad x \neq y.$$

Therefore,

$$\lim_{(x,y) \rightarrow (0,0)} \frac{f(x, y) - f(0, 0) - f_x(0, 0)x - f_y(0, 0)y}{\sqrt{x^2 + y^2}} = 0,$$

so f is differentiable at $(0, 0)$, but f_x and f_y are not continuous at $(0, 0)$.